

Supplementary Material S1: Detailed Study Characteristics (n=29)

This table provides the complete dataset of the 29 studies included in the systematic review, following the "Conservative Scenario" (n=29) which excludes non-UAV and non-agricultural studies.

Table 1: Detailed characteristics of the 29 included studies.

ID	Reference	Algorithm	Val.	Val. Type	UAVs	T	E	C	App.
S01	Chen et al. (2021)	ACO	Sim	Multi-agent	2–10	–	–	✓	Mon.
S02	Lin et al. (2022)	ABC	Sim	Framework-only	1	–	–	✓	Gen.
S03	Liu et al. (2021)	SSA	Sim	Framework-only	1	–	–	–	Gen.
S04	Liu et al. (2021)	GA+ABC	Sim	Framework-only	1–3	–	✓	✓	Gen.
S05	Liu et al. (2021)	SSA hybrid	Sim	Multi-agent	3	–	–	–	Mon.
S06	Mathew et al. (2021)	PSO+ACO	Sim	Framework-only	1	✓	–	–	Gen.
S07	Ntakolia & Lyridis (2021)	Fuzzy/SIGPAF	Sim	Framework-only	1	–	–	–	Gen.
S08	Pan et al. (2022)	SFLA	Sim	Framework-only	1–4	–	–	–	Mon.
S09	Phung & Ha (2021)	PSO	Sim	Framework-only	1	✓	✓	✓	Map.
S10	Puente-Castro et al. (2022)	Review	N/A	Review	N/A	–	–	–	Gen.
S11	Sharma et al. (2022)	Review	N/A	Review	N/A	–	–	–	Gen.
S12	Yu et al. (2021)	PSO hybrid	Sim	Framework-only	1	–	✓	✓	Gen.
S13	Chu et al. (2022)	PSO	Sim	Framework-only	1	✓	–	–	Spr.
S14	Fevgas et al. (2022)	Review	N/A	Review	N/A	–	–	–	Gen.
S15	Israr et al. (2022)	Review	N/A	Review	N/A	–	–	–	Gen.
S16	Ji et al. (2022)	PSO	Sim	Framework-only	1	–	–	–	Gen.
S17	Ait Saadi et al. (2022)	Review	N/A	Review	N/A	–	–	–	Gen.
S18	Selma et al. (2022)	ABC	Sim	Framework-only	1	–	–	✓	Gen.
S19	Shafiq et al. (2022)	ACO	Sim	Multi-agent	2	–	–	✓	Gen.
S20	Ahmed et al. (2021)	PSO	Sim	Multi-agent	4	✓	✓	✓	Gen.
S21	Xu et al. (2022)	Wolf Pack	Sim	Multi-agent	5–15	✓	–	–	Gen.
S22	Deng et al. (2023)	PSO	Sim	Framework-only	1	✓	–	–	Gen.

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ID	Reference	Algorithm	Val.	Val. Type	UAVs	T	E	C	App.
S23	Xiao et al. (2025)	NOA	Sim	Multi-agent	6–8	✓	–	–	Gen.
S24	Rao et al. (2024)	GWO	Sim	Framework-only	1	✓	–	–	Gen.
S25	Zhang & Zhang (2022)	PSO	Sim	Framework-only	2+	✓	–	✓	Gen.
S26	Zuo et al. (2025)	APF+PSO	Sim	Multi-agent	5–25	✓	–	–	Gen.
S27	Yang et al. (2025)	DOA	Sim	Framework-only	1	✓	–	–	Gen.
S28	Yang et al. (2023)	Review	N/A	Review	N/A	–	–	–	Gen.
S29	Tang et al. (2022)	Review	N/A	Review	N/A	–	–	–	Gen.